

PROVO, UT, USA

WMO: 725724

Lat: 40.219N

Lon: 111.723W

Elev: 4497

StdP: 12.46

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24174

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 8.8	(c) 13.2	(d) 2.4	(e) 7.4	(f) 10.4	(g) 7.0	(h) 9.3	(i) 15.7	(j) 24.9	(k) 33.6	(l) 21.8	(m) 33.3	(n) 2.6	(o) 140	(p) 0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.7	95.2	62.5	92.5	62.2	90.2	62.0	66.8	87.3	65.4	85.5	64.2	84.2	6.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 60.6	(b) 93.4	(c) 75.6	(d) 57.5	(e) 83.3	(f) 75.2	(g) 55.5	(h) 77.7	(i) 75.3	(j) 33.9	(k) 87.4	(l) 32.8	(m) 85.0	(n) 31.7	(o) 84.2	(p) 72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 24.2	(b) 20.2	(c) 17.5	(e) 4.7	(f) 99.8	(g) 7.6	(h) 2.3	(i) -0.8	(j) 101.5	(k) -5.2	(l) 102.8	(m) -9.4	(n) 104.1	(o) -15.0	(p) 105.7		
			DB													
			WB													
				4.7	99.8	7.6	2.3	-0.8	101.5	-5.2	102.8	-9.4	104.1	-15.0	105.7	
				3.1	70.5	6.9	1.4	-1.9	71.6	-5.9	72.4	-9.8	73.2	-14.8	74.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	29.9	34.1	43.2	49.3	58.3	67.7	76.3	74.0	63.3	50.5	38.8	30.1
	DBStd	17.46	8.36	7.83	7.05	7.16	7.80	7.11	4.54	4.05	7.50	7.39	8.11	7.77
	HDD50	2470	624	447	232	98	18	2	0	0	5	84	343	617
	HDD65	5835	1089	866	677	471	234	56	2	2	118	452	786	1082
	CDD50	2977	0	1	21	79	274	532	815	744	405	99	7	0
	CDD65	867	0	0	0	1	26	136	353	281	68	2	0	0
	CDH74	10702	0	0	2	27	441	1874	4324	3125	851	58	0	0
	CDH80	4861	0	0	0	3	129	790	2247	1417	268	7	0	0
(14) Wind	WSAvg	6.4	4.9	6.0	7.2	7.9	7.2	7.4	6.6	6.6	6.3	6.1	5.5	5.2
(15) Precipitation	PrecAvg	18.4	1.9	1.7	1.7	2.1	1.9	1.2	0.5	0.7	1.2	2.0	1.7	2.0
	PrecMax	22.8	3.8	3.3	3.3	4.6	5.5	3.7	1.3	1.5	2.5	4.6	3.1	5.3
	PrecMin	12.3	0.4	0.2	0.6	0.2	0.2	0.0	0.0	0.1	0.1	0.2	0.3	0.7
	PrecStd	3.0	1.1	0.8	0.8	1.0	1.3	0.9	0.4	0.4	0.7	1.1	0.8	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	58.7	72.0	77.9	90.0	96.3	99.8	95.9	90.5	80.8	64.7	54.4
		MCWB	42.2	45.0	50.8	54.3	61.6	60.1	64.2	62.5	60.7	55.1	47.4	43.2
	2%	DB	46.7	53.5	64.5	72.4	82.6	91.6	97.0	92.9	86.7	74.0	60.8	48.0
		MCWB	39.2	42.3	47.6	51.5	57.1	59.9	63.4	62.2	59.9	53.8	46.3	40.1
	5%	DB	44.3	49.3	60.9	68.0	79.1	89.0	93.5	90.5	82.8	69.9	55.5	44.9
		MCWB	37.9	40.4	45.9	50.3	56.2	59.9	63.0	62.3	59.6	52.9	43.8	37.9
	10%	DB	40.8	45.6	55.9	63.7	74.4	84.2	91.0	88.2	79.4	64.2	52.2	41.4
		MCWB	35.7	38.2	44.0	48.4	55.1	58.9	62.9	62.2	58.6	50.1	42.6	35.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.4	46.0	51.4	55.4	63.3	64.9	69.7	67.6	65.8	58.6	49.8	44.1
		MCDB	49.5	56.5	68.1	74.5	85.7	84.9	88.9	86.8	82.9	73.6	61.9	53.2
	2%	WB	40.4	43.2	49.1	52.8	59.7	62.7	67.5	65.8	63.3	55.4	47.2	40.8
		MCDB	46.3	51.8	63.5	69.2	78.7	84.0	89.2	86.0	80.4	70.4	58.3	47.1
	5%	WB	38.4	40.9	46.9	50.7	57.4	61.4	66.2	64.6	61.1	53.2	44.9	38.5
		MCDB	43.4	48.1	59.1	65.7	74.6	83.7	86.9	84.9	78.8	67.3	54.7	43.9
	10%	WB	36.3	38.8	44.5	49.1	55.5	60.1	65.0	63.6	59.4	51.0	43.1	36.3
		MCDB	40.8	45.0	55.4	62.8	73.0	81.9	85.1	83.7	76.6	63.9	51.1	41.3
(35) Mean Daily Temperature Range	5% DB	MDBR	14.7	16.3	20.1	21.4	24.4	27.9	29.7	28.5	26.9	23.8	18.6	14.6
		MCDBR	17.2	20.6	27.7	29.4	31.7	34.3	33.2	32.0	31.7	31.5	25.3	17.5
	5% WB	MCWBR	11.4	12.1	14.7	14.5	14.2	14.4	13.4	13.3	14.6	15.8	14.3	11.3
		MCWBR	16.0	18.9	26.1	27.8	29.0	31.2	29.1	28.2	28.6	28.8	23.5	16.6
(40) Clear-Sky Solar Irradiance	taub		0.251	0.264	0.280	0.309	0.317	0.313	0.347	0.343	0.306	0.284	0.262	0.250
		taud	2.473	2.506	2.528	2.455	2.493	2.537	2.473	2.480	2.564	2.598	2.587	2.507
	Ebn at Noon		297	307	312	306	304	304	292	292	298	296	292	289
		Edh at Noon	24	27	30	34	34	32	34	33	29	24	21	21
(44) All-Sky Solar Radiation	RadAvg		707	970	1421	1777	2083	2473	2331	2031	1705	1196	791	589
	RadStd		61	95	100	114	175	192	115	108	124	98	80	66

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page